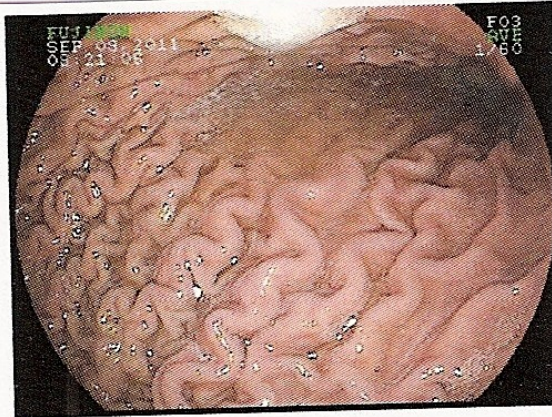

Module 13
Gastrointestinal
system

BIO262
Physiology



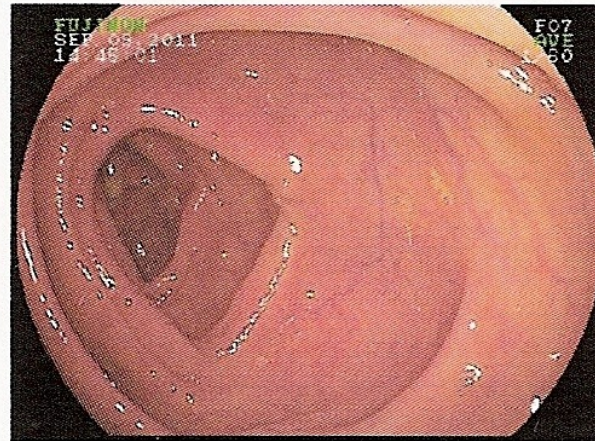
3 Lower Third of the Esophagus: Normal



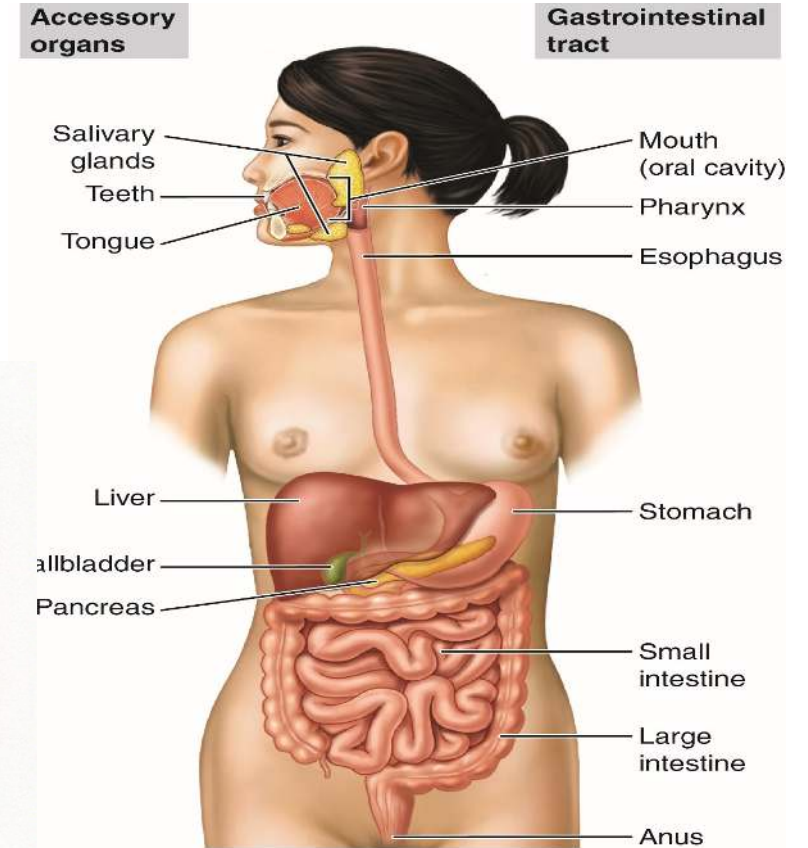
4 Gastric Body: Normal



11 Duodenal Bulb: Normal



9 Descending Colon: Exam Otherwise Normal



Digestion

- ✓ Ingestion
 - ✓ Mastication
 - ✓ Propulsion – Peristalsis, Segmentation
 - ✓ Mechanical digestion – Chewing, churning, bile
 - ✓ Chemical digestion – Digestive enzymes
 - ✓ Deglutination
 - ✓ Absorption
 - ✓ Defecation
-

Stomach

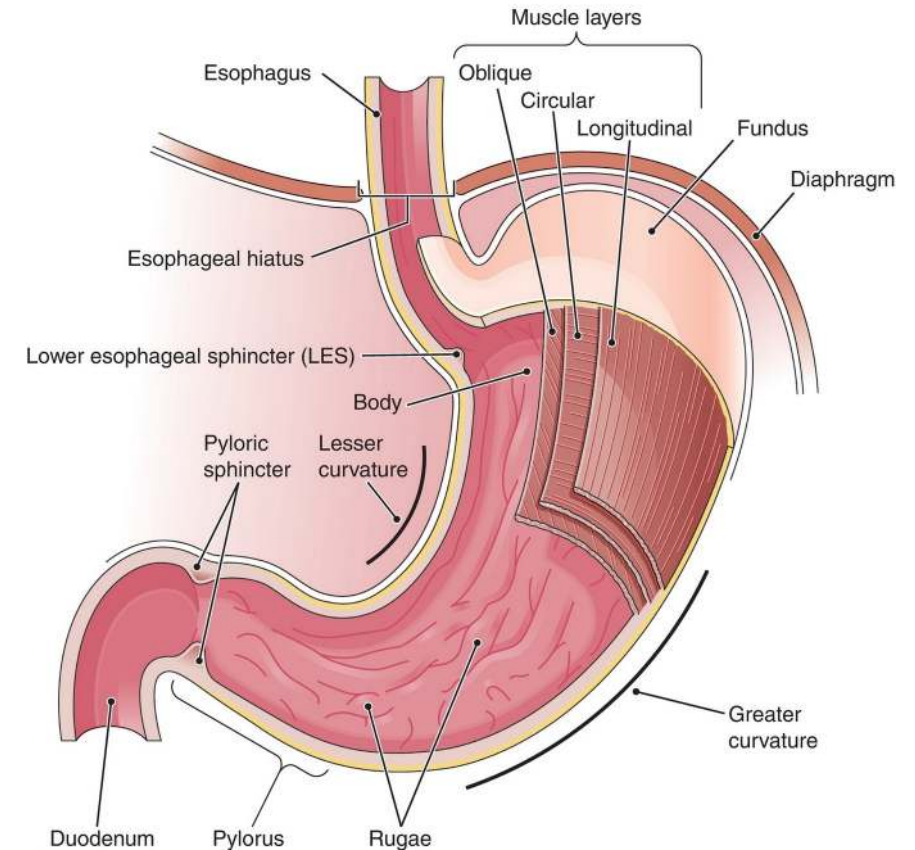
Three parts: Fundus, Body, Antrum

Function: mixes, stores food – goal is CHYME

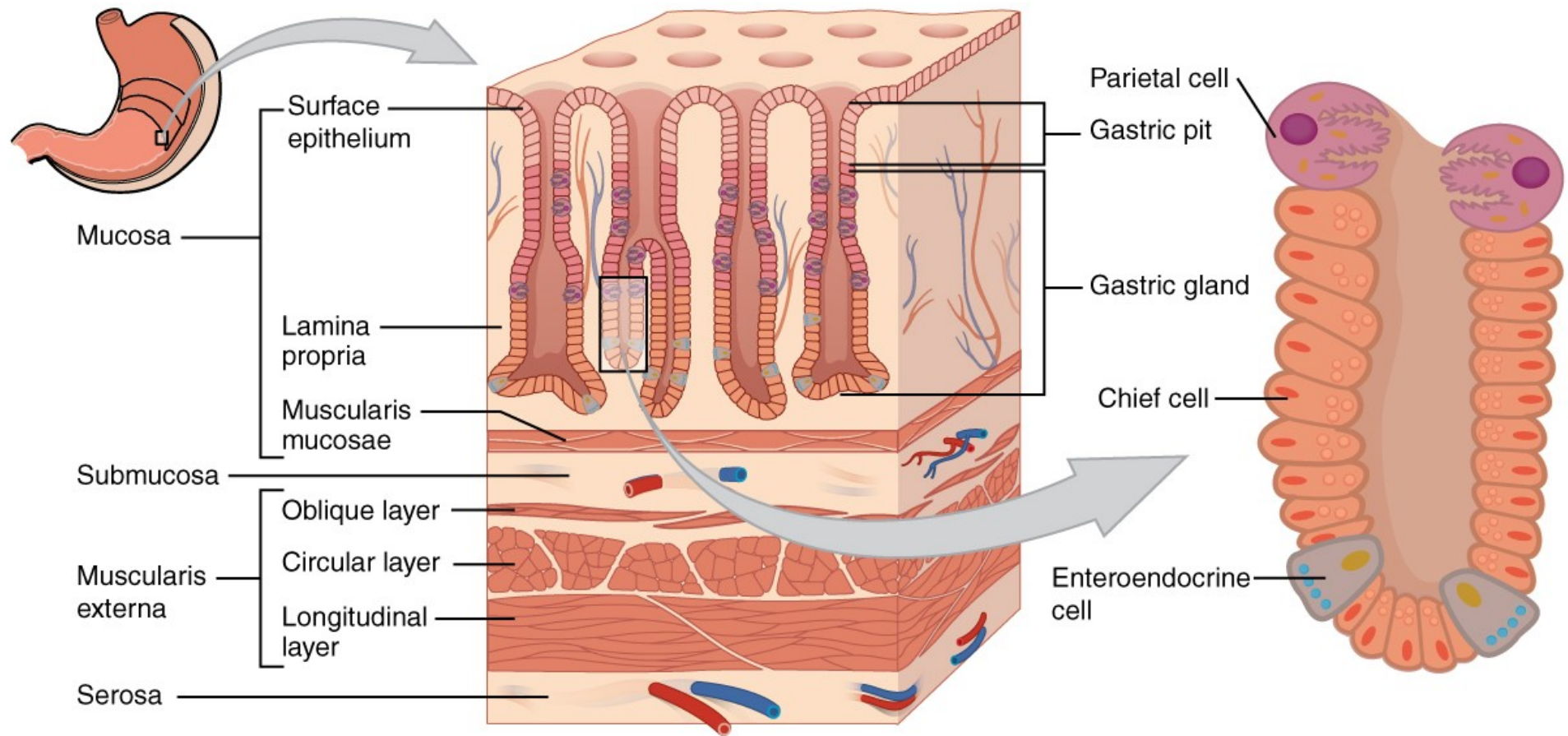
pH of contents ranges from 2-5

Aspirin, alcohol, electrolytes are absorbed

Pyloric Sphincter opens when content consistency is CHYME – dumps into duodenum.



Stomach Mucosa

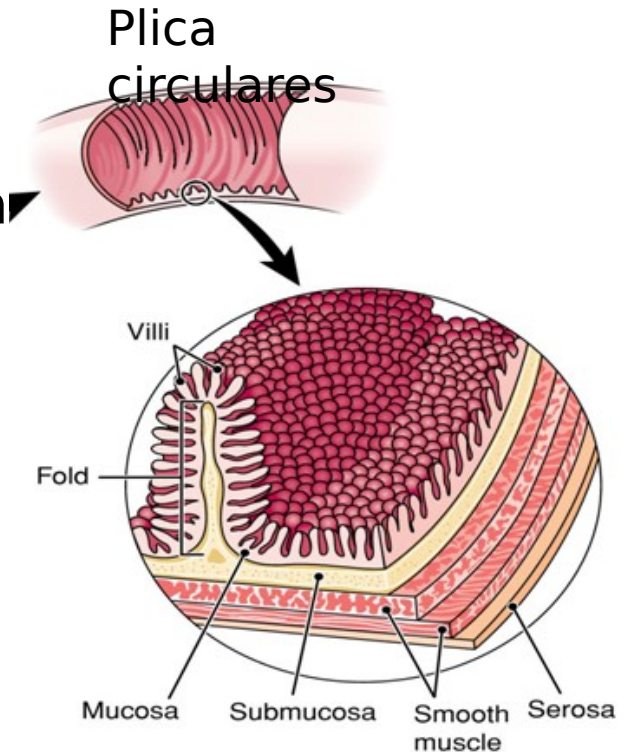


Stomach Cells:

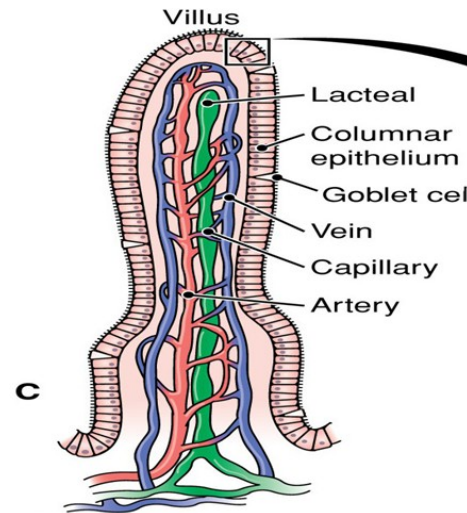
- Chief: Pepsinogen/pepsin digests proteins
 - Parietal: Hydrochloric acid, water, intrinsic factor
 - Goblet: protection – mucus – these cells are throughout the intestinal tract also but in lesser numbers
 - G-Cells – secretes hormone: gastrin, stimulating chief and parietal cells
-

Duodenum, Jejunum, Ileum

- Function: digestion , absorption
 - Functioning unit: Villi/microvilli/Brush border
- Production/release of Digestive Enzymes:
 - Peptidases - **protein** digestion
 - Maltase and sucrase - **maltose/sucrose** digestion
 - Lactase - **lactose** (milk/dairy) digestion
 - ****Secretin and Cholecystikin****



11 Duodenal Bulb: Normal



Pancreas



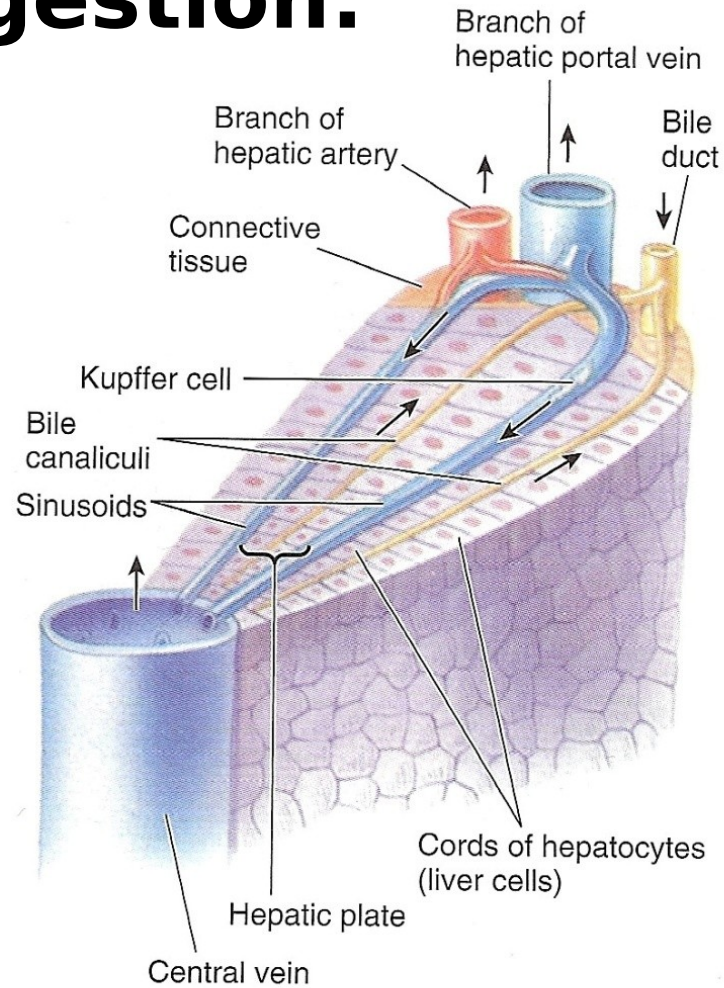
What do these tell the pancreas to release?

➤ **Secretin:**

➤ **Cholecystokinin:**

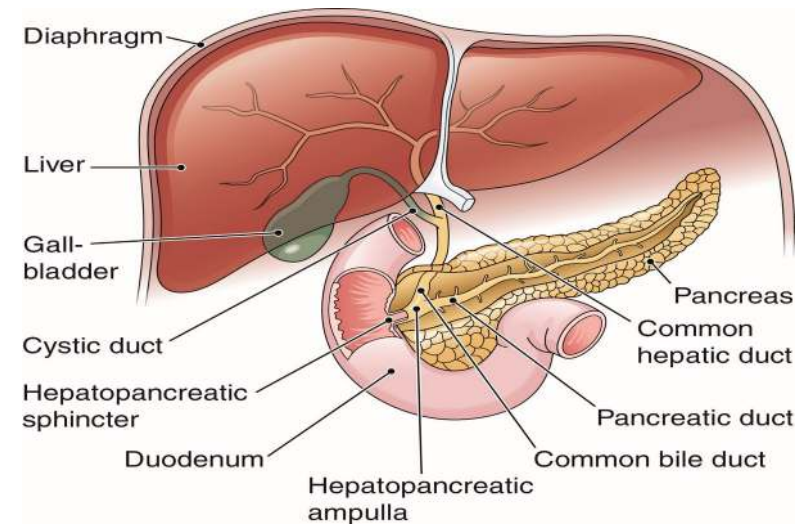
- Beta cells produce insulin – why?
- **acini**—*exocrine glands* of the pancreas producing *enzymes* – why?
- **bicarbonate juice**—neutralizing the acidic pH of the stomach contents entering the duodenum
- Enzymes:
 - Amylase – **carbohydrate/starch** digestion
 - Lipase - **fats** digestion
 - Nucleases – RNA/DNA
 - Proteolytic & trypsin - proteins

Liver and digestion:

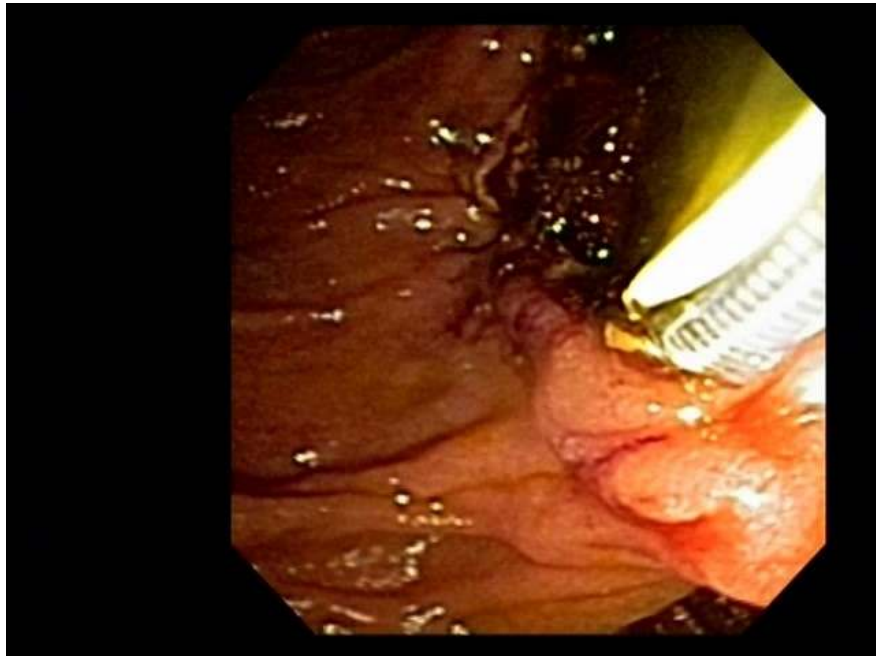


(b) Wedge of a hepatic lobule

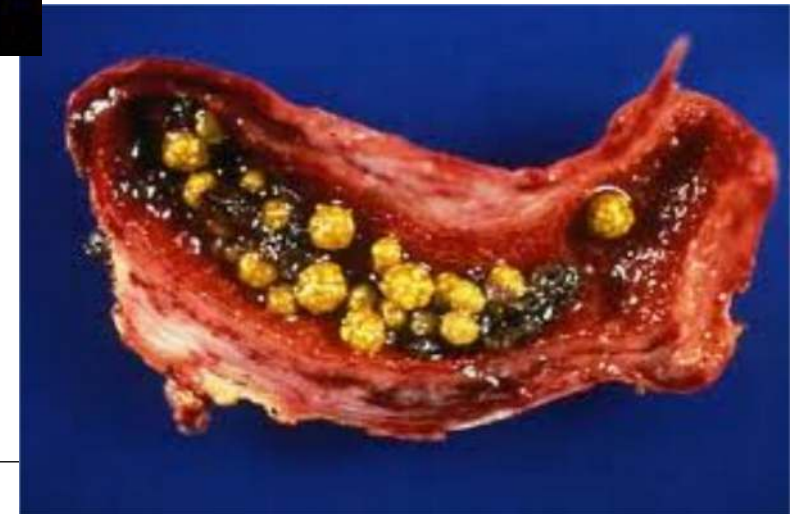
- 1- Kupffer cells A - liver cell
- 2- Bile B - stores iron, lipids, heavy metals - is a phagocyte
- 3- Hepatocyte C - carries bilirubin, cholesterol and bile salts. Emulsifies fats..

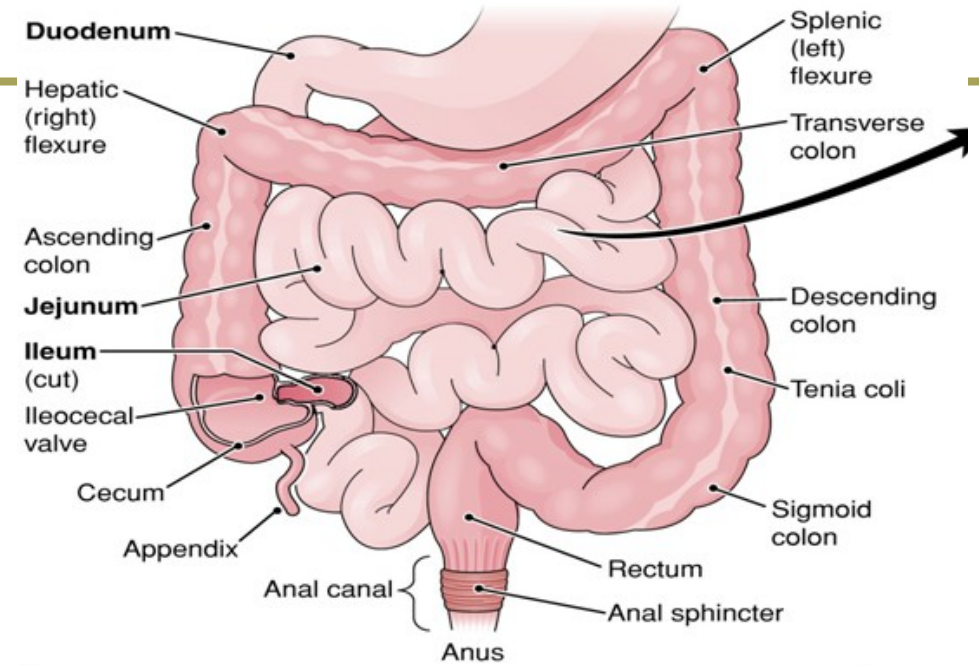
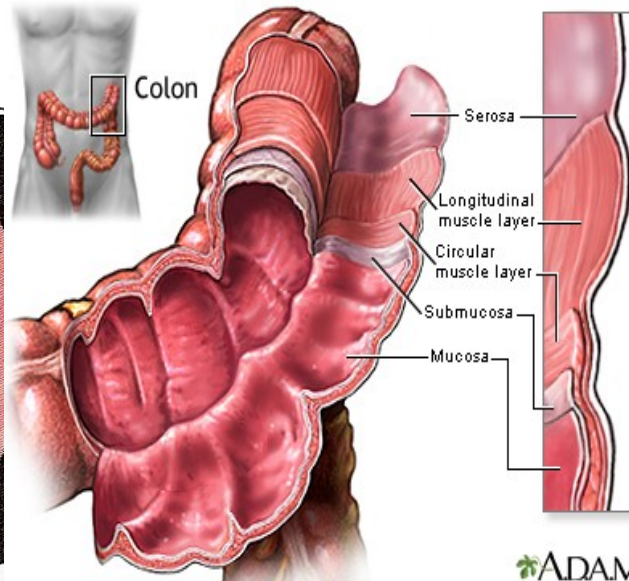
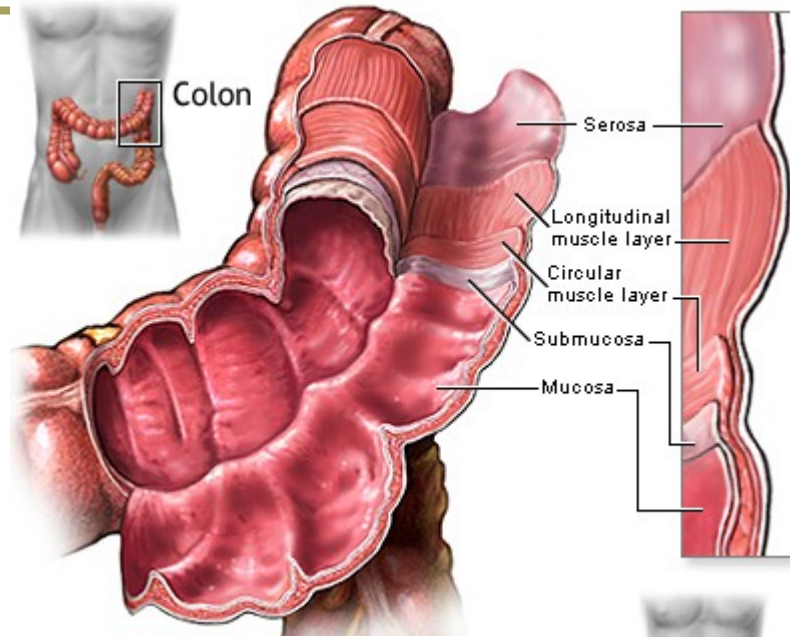


Gallbladder - BILE!!!!



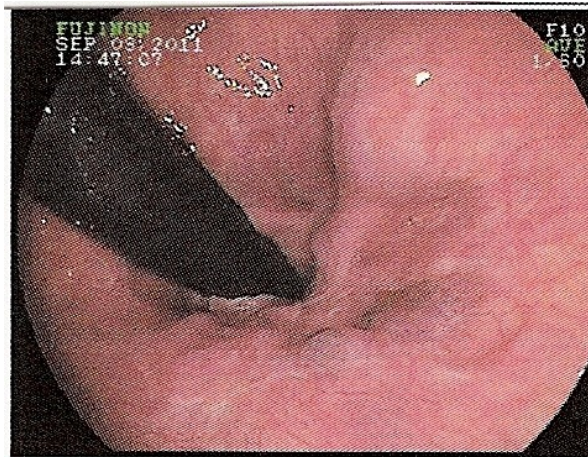
Gallbladder stores and concentrates bile.
CCK causes it to contract and release bile into the duodenum.....





A

B



6 Rectum: Hemorrhoids

Large Intestine = Colon
Absorption of water and vitamins only.
Forms fecal matter from undigestible wastes (fiber....)

Matching

- | | |
|------------|---|
| 1- pepsin | A – digests polypeptides (proteins) into AA |
| 2- sucrase | B – digests proteins into smaller polypeptides (in the stomach) |
| 3- amylase | C – Secreted by the pancreas into duodenum neutralizing acid |
| 4- gastrin | |
| 5- bicarb | D – breaks down sugars – released by small intestines |
| 6- trypsin | E – breaks down starch – released by salivary glands & pancreas |
| | F – stimulates parietal/chief cells to release gastric juices |
-

Terminology – matching!

1. Ingestion
2. Mastication
3. Propulsion
4. Peristalsis
5. Segmentation
6. Mechanical Digestion
7. Chemical Digestion
8. Deglutition
9. Absorption
10. Defecation

- A. Swallowing
 - B. Food entering
 - C. Squeezing food along intestines
 - D. Nutrients move into blood
 - E. Bowel movement
 - F. Enzymes break down food
 - G. Mixing and churning of food chunks
 - H. Breaks chunks into smaller pieces
 - I. Chewing
 - J. Swallowing and peristalsis
-

Reminders

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• **DUE WEDNESDAY:** 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
 - Module 11 quiz (Respondus)
 - Peer response in discussion
 - Lab quiz (Respondus)
-